

Pulmonary arterial hypertension patients with concomitant left ventricular failure

Treatment with diuretics should be considered in patients with evidence of fluid retention before the start of treatment with Bosentan.

Pulmonary arterial hypertension associated with HIV infection

An increased long-term risk of hepatic toxicity and hematological adverse events cannot be excluded when Bosentan is used in combination with antiretroviral medicinal products. Due to the potential for interactions related to the inducing effect of Bosentan on CYP450, which could affect the efficacy of antiretroviral therapy, these patients should also be monitored carefully regarding their HIV infection.

Pulmonary hypertension secondary to chronic obstructive pulmonary disease (COPD)

An increase in minute ventilation and a decrease in oxygen saturation were observed, and the most frequent adverse event was dyspnea, which resolved with discontinuation of Bosentan. Concomitant use with other medicinal products

Concomitant use of Bosentan and Cyclosporine A is contraindicated.

Concomitant use of Bosentan with Glibenclamide, Fluconazole and Rifampicin is not recommended.

Concomitant administration of both a CYP3A4 inhibitor and a CYP2C9 inhibitor with Bosentan should be avoided.

✧ **PREGNANCY AND LACTATION:**

Pregnancy

There are no reliable data on the use of Bosentan in pregnant women. The potential risk for humans is still unknown. Bosentan is contraindicated in pregnancy.

Women of childbearing potential

Before the initiation of Bosentan treatment in women of childbearing potential, the absence of pregnancy should be checked, appropriate advice on reliable methods of contraception provided, and reliable contraception initiated. Patients and prescribers must be aware that due to potential pharmacokinetic interactions, Bosentan may render hormonal contraceptives ineffective. Therefore, women of childbearing potential must not use hormonal contraceptives (including oral, injectable, transdermal or implantable forms) as the sole method of contraception but must use an additional or an alternative reliable method of contraception. If there is any doubt about what contraceptive advice should be given to the individual patient, consultation with a gynecologist is recommended. Because of possible hormonal contraception failure during Bosentan treatment, and also bearing in mind the risk that pulmonary hypertension severely deteriorates with pregnancy, monthly pregnancy tests during treatment with Bosentan are recommended to allow early detection of pregnancy.

Breastfeeding

It is not known whether Bosentan is excreted into human breast milk. Breastfeeding is not recommended during treatment with Bosentan.

Fertility

It cannot be excluded that Bosentan may have a detrimental effect on spermatogenesis in men. In male children, a long-term impact on fertility after treatment with Bosentan cannot be excluded.

✧ **SIDE EFFECT:**

Tabulated list of adverse reactions:

System Organ Class	Frequency	Adverse reaction
Blood and lymphatic disorders	Common	Anemia. Hemoglobin decrease
	Not known	Anemia or hemoglobin decreases requiring blood cells transfusion
	Uncommon	Thrombocytopenia
	Uncommon	Neutropenia, leukopenia
Immune system disorders	Common	Hypersensitivity reactions (including dermatitis, pruritus and rash)
	Rare	Anaphylaxis and/or angioedema
Nervous system disorders	Very common	Headache
	Common	Syncope
Eye disorders	Not known	Blurred vision
Cardiac disorders	Common	Palpitations
Vascular disorders	Common	Flushing
	Common	Hypotension
Respiratory, thoracic and mediastinal disorders	Common	Nasal congestion
Gastrointestinal disorders	Common	Gastroesophageal reflux disease, diarrhea
Hepato-biliary disorders	Very common	Abnormal liver function test
	Uncommon	Aminotransferase elevations associated with hepatitis (including possible exacerbation of underlying hepatitis) and/or jaundice
	Rare	Liver cirrhosis, liver failure
Skin and subcutaneous disorders	Common	Erythema
General disorders and administration site conditions	Very common	Edema, fluid retention

Pediatric population

Liver test abnormalities and decrease hemoglobin concentration was reported.

✧ **EFFECTS ON ABILITY TO DRIVE AND USE MACHINE:**

No specific studies have been conducted to assess the direct effect of Bosentan on the ability to drive and use machines. However, Bosentan may induce hypotension, with symptoms of dizziness, blurred vision or syncope that could affect the ability to drive or use machines.

✧ **CONTRAINDICATION:**

- Hypersensitivity to the active substance or to any of the excipients
- Moderate to severe hepatic impairment, i.e. Child-Pugh class B or C
- Baseline values of liver aminotransferases, i.e. aspartate aminotransferase (AST) and/or alanine aminotransferase (ALT), greater than 3 × the upper limit of normal (ULN)
- Concomitant use of Cyclosporine A
- Pregnancy
- Women of childbearing potential who are not using reliable methods of contraception

✧ **DRUG INTERACTION:**

Bosentan is an inducer of the cytochrome P450 (CYP) isoenzymes CYP2C9 and CYP3A4. Consequently, plasma concentrations of substances metabolized by these isoenzymes will be decreased when Bosentan is co-administered. The possibility of altered efficacy of medicinal products metabolized by these isoenzymes should be considered. The dosage of these products may need to be adjusted after initiation, dose change or discontinuation of concomitant Bosentan treatment.

Bosentan is metabolized by CYP2C9 and CYP3A4. Inhibition of these isoenzymes may increase the plasma concentration of Bosentan (see Ketoconazole). The influence of CYP2C9 inhibitors on Bosentan concentration has not been studied. The combination should be used with caution.

Fluconazole and other inhibitors of both CYP2C9 and CYP3A4

Co-administration with Fluconazole, which inhibits mainly CYP2C9, but to some extent also CYP3A4, could lead to large increases in plasma concentrations of Bosentan. The combination is not recommended. For the same reason, concomitant administration of both a potent CYP3A4 inhibitor (such as Ketoconazole, Itraconazole or Ritonavir) and a CYP2C9 inhibitor (such as Voriconazole) with Bosentan is not recommended.

Cyclosporine A

Co-administration of Bosentan and cyclosporine A (a calcineurin inhibitor) is contraindicated (see Contraindication). When co-administered, initial trough concentrations of Bosentan were approximately 30-fold higher than those measured after Bosentan alone. At steady state, Bosentan plasma concentrations were 3- to 4-fold higher than with Bosentan alone. The mechanism of this interaction is most likely inhibition of transport protein-mediated uptake of Bosentan into hepatocytes by cyclosporine. The blood concentrations of Cyclosporine A (a CYP3A4 substrate) decreased by approximately 50%. This is most likely due to induction of CYP3A4 by Bosentan.

Tacrolimus, Sirolimus

Co-administration of Tacrolimus or Sirolimus and Bosentan has not been studied in man but co-administration of Tacrolimus or Sirolimus and Bosentan may result in increased plasma concentrations of Bosentan in analogy to co-administration with cyclosporine A. Concomitant Bosentan may reduce the plasma concentrations of Tacrolimus and Sirolimus. Therefore, concomitant use of Bosentan and Tacrolimus or Sirolimus is not advisable. Patients in need of the combination should be closely monitored for adverse events related to Bosentan and for Tacrolimus and Sirolimus blood concentrations.

Glibenclamide

Co-administration of Bosentan 125 mg twice daily for 5 days decreased the plasma concentrations of Glibenclamide (a CYP3A4 substrate) by 40%, with potential significant decrease of the hypoglycemic effect. The plasma concentrations of Bosentan were also decreased by 29%. In addition, an increased incidence of elevated aminotransferase may be observed in patients receiving concomitant therapy. Both Glibenclamide and Bosentan inhibit the bile salt export pump, which could explain the elevated aminotransferases. This combination should not be used. No drug-drug interaction data are available with the other sulfonylureas.

Rifampicin

Co-administration of Bosentan 125 mg twice daily with Rifampicin, a potent inducer of CYP2C9 and CYP3A4 for 7 days, will decrease the plasma concentrations of Bosentan by 58%, and this decrease could achieve almost 90% in an individual case. As a result, a significantly reduced effect of Bosentan is expected when it is co-administered with Rifampicin. Concomitant use of Rifampicin and Bosentan is not recommended. Data on other CYP3A4 inducers, e.g. Carbamazepine, Phenobarbital, Phenytoin and St. John's Wort are lacking, but their concomitant administration is expected to lead to reduced systemic exposure to Bosentan. A clinically significant reduction of efficacy cannot be excluded.

Lopinavir + Ritonavir (and other Ritonavir-boosted protease inhibitors)

Co-administration of Bosentan 125 mg twice daily and Lopinavir + Ritonavir 400 + 100 mg twice daily for 9.5 days will result in initial trough plasma concentrations of Bosentan that were approximately 48-fold higher than those measured after Bosentan administered alone. On day 9, plasma concentrations of Bosentan were approximately 5-fold higher than with Bosentan administered alone. Inhibition by Ritonavir of transport protein-mediated uptake into hepatocytes and of CYP3A4, thereby reducing the clearance of Bosentan, most likely causes this interaction. When administered concomitantly with Lopinavir + Ritonavir, or other Ritonavir-boosted protease inhibitors, the patient's tolerability of Bosentan should be monitored.

After co-administration of Bosentan for 9.5 days, the plasma exposures to Lopinavir and Ritonavir may be decreased to a clinically non-significant extent (by approximately 14% and 17%, respectively). However, full induction by Bosentan might not have been reached and a further decrease of protease inhibitors cannot be excluded. Appropriate monitoring of the HIV therapy is recommended. Similar effects would be expected with other Ritonavir-boosted protease inhibitors (see Precaution)

Other antiretroviral agents

No specific recommendation can be made with regard to other available antiretroviral agents due to the lack of data. Due to the marked hepatotoxicity of Nevirapine, which could add to Bosentan liver toxicity, this combination is not recommended.

Hormonal contraceptives

Co-administration of Bosentan 125 mg twice daily for 7 days with a single dose of oral contraceptive containing Norethisterone 1 mg + Ethinyl estradiol 35 mcg decreased the AUC of Norethisterone and

Ethinyl estradiol by 14% and 31%, respectively. However, decreases in exposure were as much as 56% and 66%, respectively, in individual subjects. Therefore, hormone-based contraceptives alone, regardless of the route of administration (i.e. oral, injectable, transdermal or implantable forms), are not considered as reliable methods of contraception (see Precaution and Pregnancy and lactation)

Warfarin

Co-administration of Bosentan 500 mg twice daily for 6 days may decrease the plasma concentrations of both S-Warfarin (a CYP2C9 substrate) and R-warfarin (a CYP3A4 substrate) by 29% and 38%, respectively. Concomitant administration of Bosentan with Warfarin in patients with PAH may not result in clinically relevant changes in International Normalized Ratio (INR) or Warfarin dose. No dose adjustment is needed for Warfarin and similar oral anticoagulant agents when Bosentan is initiated, but intensified monitoring of INR is recommended, especially during Bosentan initiation and the up-titration period.

Simvastatin

Co-administration of Bosentan 125 mg twice daily for 5 days decreased the plasma concentrations of Simvastatin (a CYP3A4 substrate) and its active β -hydroxy acid metabolite by 34% and 46%, respectively. The plasma concentrations of Bosentan were not affected by concomitant Simvastatin. Monitoring of cholesterol levels and subsequent dosage adjustment should be considered.

Ketoconazole

Co-administration for 6 days of Bosentan 62.5 mg twice daily with Ketoconazole, a potent CYP3A4 inhibitor, increased the plasma concentrations of Bosentan approximately 2-fold. No dose adjustment of Bosentan is considered necessary. Although not demonstrated through *in vivo* studies, similar increases in Bosentan plasma concentrations are expected with the other potent CYP3A4 inhibitors (such as Itraconazole or Ritonavir). However, when combined with a CYP3A4 inhibitor, patients who are poor metabolizers of CYP2C9 are at risk of increases in Bosentan plasma concentrations that may be of higher magnitude, thus leading to potential harmful adverse events.

Epoprostenol

Combination of Bosentan and Epoprostenol may indicate that after both single- and multiple-dose administration, the C_{max} and AUC values of Bosentan is similar in patients with or without continuous infusion of Epoprostenol (see Pharmacology).

Sildenafil

Co-administration of Bosentan 125 mg twice daily (steady state) with Sildenafil 80 mg three times a day (at steady state) concomitantly administered during 6 days may result in a 63% decrease in the Sildenafil AUC and a 50% increase in the Bosentan AUC. Caution is recommended in the case of co-administration.

Tadalafil

Bosentan (125 mg twice daily) reduced Tadalafil (40 mg once per day) systemic exposure by 42% and C_{max} by 27% following multiple dose co-administrations. Tadalafil did not affect the exposure (AUC and C_{max}) of Bosentan or its metabolites.

Digoxin

Co-administration for 7 days of Bosentan 500 mg twice daily with Digoxin decreased the AUC, C_{max} and C_{min} of Digoxin by 12%, 9% and 23%, respectively. The mechanism for this interaction may be induction of P-glycoprotein. This interaction is unlikely to be of clinical relevance.

Pediatric population

Interaction studies have only been performed in adults.

✧ **OVERDOSE AND TREATMENT:**

The most common adverse reaction of Bosentan when administered as a single dose of up to 2,400 mg for 2 months in patients with a disease other than pulmonary hypertension was headache of mild to moderate intensity.

Massive overdose may result in pronounced hypotension requiring active cardiovascular support. An overdose of 10,000 mg of Bosentan may experience symptoms of nausea, vomiting, hypotension, dizziness, sweating and blurred vision. **Note:** Bosentan is not removed through dialysis.

✧ **STORAGE:**

Store at temperature of not more than 30°C.

✧ **DOSAGE FORM AND PACKAGING AVAILABLE:**

125 mg Tablet, Blister 1x10's, 6x10's

✧ **DATE OF REVISION:**

August 18, 2023

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomchoijorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

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Approved by: (MD, Entry Patient and Close)				
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